11 Saint James Street
Petrie Terrace, Qld 4000
T +61 7 3831 4644
info@m3architecture.com

m3architecture

PRELIMINARY
NOT FOR CONSTRUCTION

Stanwell

Stanwell New Projects Office
Switchyard Road,
Stanwell, QLD, 4702

25010 / cd Spec D / T1

Appendix D - Waterproofing Specification

Revision	Date	Description	Issued by
T1	11/03/2026	For Tender	JB

Tremco CPG Australia Pty Ltd
63 Radley Street,
Virginia Queensland 4014 Australia
T. (07) 3889 9222
www.tremcocpg.com.au

13 October 2025

M3 Architecture
11 St James Street
Petrie Terrace, QLD 4000

Attention: Jaydn Bowe
Subject: Tremco CPG Product Suggestions
Project: Stanwell Power

Dear Jaydn,

Thank you for the opportunity to recommend Tremco CPG systems for the above-mentioned project. With this in mind, we proudly submit the following product suggestions for your consideration.

Introduction to Tremco CPG

At Tremco Construction Products Group (CPG), we're committed to shaping the future through innovative approaches and sustainable solutions. From joint sealing, façade bonding and insulation through to passive fire protection, repair mortars and grouts, performance resin flooring, waterproofing and roofing solutions- our broad range makes us the ideal partner to solve the complex challenges faced by today's specification experts.

Design and Specification Assistance

Tremco CPG are committed to assisting project stakeholders with the design and specification of the nominated Tremco CPG systems. We are available to attend design workshops, to gain a more comprehensive understanding of the project specifics, to enable us to tailor our recommendations to meet the project specific requirements. We also offer a suite of standard detail drawings, which may be subsequently used by the project design team, to develop a holistic design in accordance with the relevant regulatory requirements.

Quality Assurance Programme

Tremco CPG uses the iAuditor system to undertake periodic site observations. This service is offered as a means to mitigate against the number and severity of product application defects that occur in practice.

A representative from Tremco CPG will attend site upon request (ideally to a pre-agreed inspection schedule or ad-hoc with appropriate notice provided) to observe aspects pertaining to the physical application of the nominated systems. Following Tremco CPG's site visit, an observation report will be issued, documenting the observations made at the time of the site visit as well as any corrective actions required by the installer to achieve a complete and warrantable system. Completed in real time, all vested parties (applicator, builder, specifier etc.) have access to works as they are being observed by Tremco CPG.

Training Programme and Formal CPD Presentations

Tremco CPG are able to provide a variety of training modules (practical and/ or theory) specifically tailored to both design and construction professionals. Training and presentations may be undertaken either at a Tremco CPG premises, your offices or site.

Product Schedule

A summary of Tremco CPG's recommended products is provided in the table below, further information on the nominated systems is available via Section 5.0 of Tremco CPG's Product Specification. This product schedule shall be read in conjunction with the complete Tremco CPG product specification provided and all other applicable technical literature, available via the relevant Tremco CPG website.

Tremco CPG Product Suggestions			
Code	**Area**	**System**	**Materials Warranty**
-	Internal Wet Areas-Direct Stick Flooring	**Above, Below Screed & Walls** Vulkem H2 Plus	15-Year
-	Window & Door Reveals	Vulkem H2 Plus	10-Year
	External Deck Paver-Direct Stick	Vulkem H2 Plus	10-Year

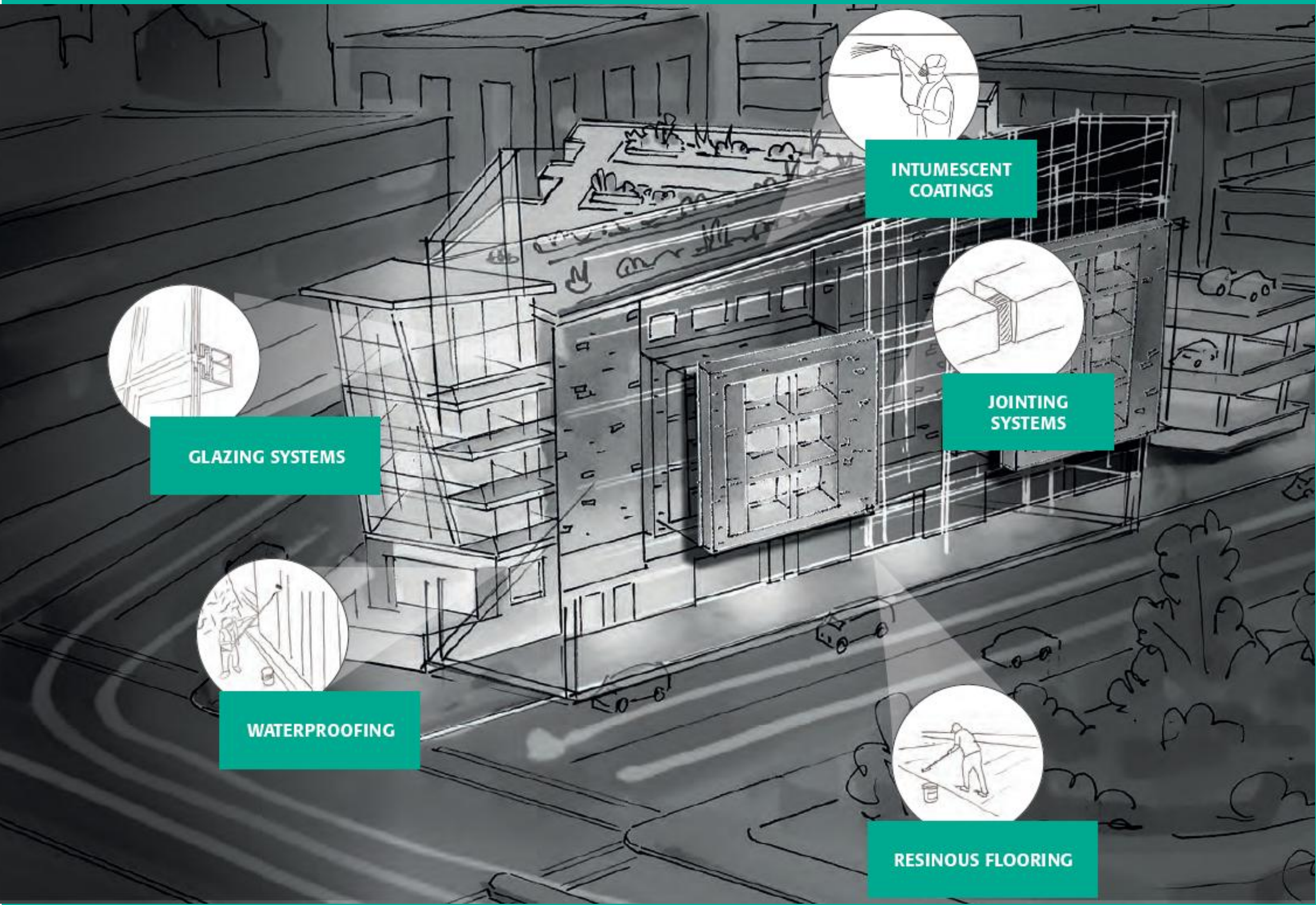
Always consider:

- Waterproofing represents approximately 1-2% of the initial cost of construction, with leaks and subsequent damage representing approximately 80% of all building defects. Make an educated decision with your product specification.
- Flooring is the only trafficable element in a building. It is constantly under stress so specifying the correct flooring system is imperative for long term success of the project.

Please do not hesitate to get in contact for any further information.

For and on behalf of Tremco CPG Australia,
Justin Allingham
Specification Manager-QLD
M. +61 497 777 747
Justin.allingham@tremco.com.au

Tremco CPG Product Specification



Prepared For: M3 Architecture
Project Details: Stanwell Power
Date: 13 October 2025
Revision: 1
Prepared By: **Justin Allingham**

Table of Contents

1. Scope

This document is applicable to the design and installation of Tremco CPG systems. All text contained within this document is provided as guidance by Tremco CPG Australia for developing the appropriate work section and relevant construction documentation.

This document shall be read in conjunction with the information contained within the relevant project specific deign and specification documentation. The extent of product application shall be as denoted on the relevant Tremco CPG technical literature, project specific design and specification documentation, and the relevant regulatory requirements. Where any contradictory information exists, adopt the most stringent requirements.

While every reasonable effort has been made to ensure that the information contained within this document is accurate, the user of this document remains solely responsible for the specification of all methods, materials, and practices.

1.1. Company Contacts

The materials nominated within this document shall be supplied by Tremco CPG Australia.
www.tremcocpg.com.au

Head Office:	1800 318 038
NSW:	(02) 9638 2755
QLD/NT:	(07) 3889 9222
VIC/SA/TAS:	(03) 9799 3900
WA:	1800 318 038

1.2. Products

All systems shall be designed and installed in strict accordance with the information contained within Tremco CPG's technical literature, including, but not necessarily limited to; material storage, substrate preparation, priming, mixing, application, curing and overcoating instructions. Products must not be altered in anyway or be used in a manner outside of the recommendations made within Tremco CPG's published technical literature, without prior written consent from Tremco CPG. Failure to adhere to this may result in voiding the offered materials warranty.

All product technical literature (including, but not limited to; Product Data Sheets, System Data Sheets and Safety Data Sheets) can be found at www.tremcocpg.com.au or by clicking on the product name, as listed within this project specification. For any other documentation required, please contact your local Tremco CPG representative.

1.3. Regulatory Requirements

All applicable regulatory requirements, including, National Construction Code, Australian Standards and Work Health Safety requirements shall be strictly adhered to in the design and installation of the nominated Tremco CPG systems.

Relevant applicable standards and codes may include, but are not necessarily limited to:

- National Construction Code- Building Code of Australia- Volume 1
- National Construction Code- Building Code of Australia- Volume 2
- Australia Building Codes Board- Housing Provision Standards
- AS 4858- Wet area membranes
- AS 3740- Waterproofing of domestic wet areas
- AS 4654.1- Waterproofing membranes for external above-ground use- materials
- AS 4654.2- Waterproofing membranes for external above-ground use- design and installation

- BS 8102- Protection of below ground structures against water ingress.
- AS 3958.1- Ceramic tiles- Guide to the installation of ceramic tiles
- AS 1288- Glass in buildings- selection and installation
- AS 2047- Windows and external glazed doors in buildings
- AS 1530.4- Fire-resistance test of elements of construction
- AS 9239.1- Reaction to fire tests for floorings- determination of the burning behaviour using a radiant heat source.
- EN IE 61340-5-1- Protection of electronic devices from electrostatic phenomena- General requirements
- HACCP- Food Safety Program
- HB 197- An introductory guide to the slip resistance of pedestrian surfaces materials
- HB 198- Guide to the specification and testing of slip resistance of pedestrian surfaces
- AS 4586- Slip resistance classification of new pedestrian surface materials
- AS 4020- Testing of products for use in contact with drinking water
- SA HB 84- Guide to concrete repair and protection
- EN 1504-3- Products and systems for the protection and repair of concrete structures
- AS 3799- Liquid membrane-forming curing compounds for concrete

Tremco CPG are committed to supplying fully compliant systems, we strive to ensure that all relevant products and systems have been independently tested and/or certified as compliant with the applicable regulatory standard, our product test reports are available upon request.

2. Substrate Preparation

Tremco CPG’s range of waterproofing, protective coating, sealant and jointing systems are typically able to be applied to a variety of stable substrates, including reinforced concrete, blockwork and select lightweight sheeting. Consult with your local Tremco CPG representative for project specific advice where alternate substrates are to be used.

Unless otherwise noted on the product specific technical literature, the following methodology shall be adopted to prepare the substrate prior to the installation of the nominated Tremco CPG waterproofing or protective coating system.

2.1. Substrate Preparation- Cementitious Substrates

- The substrate shall be appropriately cured and attain a 20 MPa minimum compressive strength. Where the membrane or coating system is to be directly trafficked and for all Flowcrete systems, the substrate shall attain a 25 MPa minimum compressive strength.
- The moisture content in the cementitious substrate shall be measured to be satisfactorily dry. Typically, unless otherwise noted on the product technical literature, the following limits are considered acceptable:
    - Relative humidity in-situ probe test, as per ASTM F2170 (preferred method for Flowcrete systems)
      < 75% RH
    - Non-destructive comparative surface moisture content, as per ASTM F2659
      < 4.5%

  Note: care should be taken where relying on the non-destructive comparative surface moisture content to verify the substrate moisture content where the element is subject to single sided drying, for example, structures where permanent steel formwork has been used, slab on grade elements or where a below screed membrane has been used. Tremco CPG typically recommends that a relative humidity in-situ probe test is undertaken in these instances.
- Slab on grade elements shall have an effective damp proof membrane in place.

- Depending on construction methodology and job site location, additional substrate testing may be required. Consult with your local Tremco CPG representative for project specific advice once the site has been established.
- The substrate shall be properly cleaned so that the surface to receive the coating, sealant or membrane is free of mould, paint, sealers, coatings, curing agents, loose particles, and other contamination or foreign matter that may interfere with the adhesion.
- The substrate shall be free of any laitance which may inhibit sufficient adhesion. Removal of laitance can be achieved through a variety of physical abrasion methods, such as, shot blasting (preferred method), sandblasting or grinding.
- Unless otherwise noted on the product specific technical literature, a CSP 3 (in line with ICRI's Technical Guideline No. 03732) shall be achieved as part of the substrate preparation process.
- Surfaces shall be made free of defects that may telegraph and show through the finished coating. All local protrusions shall be appropriately removed, and all local voids and indentations greater than 5mm shall be treated with a compatible filling compound. Consult with your local Tremco CPG representative for project specific advice regarding the recommended treatment.
- All spalled areas shall be appropriately prepared, to ensure that the substrate is clean and sound prior to membrane/ coating installation, in line with the requirements listed on the relevant product technical literature. As site specific conditions may vary, it is recommended that you contact your local Tremco CPG representative for project specific advice regarding the treatment of the spalled areas. Depending on the substrate and depth of the spalled areas, a Eucocrete or Flowcrete repair product will be recommended as the best method of repair.
- In the event of exposed reinforcing steel, it is recommended that the structural engineer of record be contacted for investigation and subsequent advice regarding the repair methodology.
- Where a screed is to be installed, in line with best practice Tremco CPG recommends that a polyethylene joint filler is installed around the perimeter of the screed to create a joint, which can then be treated with an approved compound, in line with the instructions provided below.
- Where third party engineered products or admixtures form part of the cementitious substrate to be coated, seek project specific advice from Tremco CPG to ensure that there will be no detrimental impact to the performance of the proposed Tremco CPG system.

2.2. Substrate Preparation- Lightweight Sheeting

- Select Tremco CPG waterproofing systems (excluding Flowcrete systems) may be applied over substrates constructed from lightweight sheeting. Consult with your local Tremco CPG representative for project specific advice.
- The surface to be coated must be dry, clean, smooth, firm, free of release agents, dust, mud, wires, fins, metal, projections, or any other substance that may prevent the nominated membrane system from achieving satisfactory adhesion.
- Ensure that the sheeting is appropriately installed in line with the manufacturer's and/or engineer's recommendations. Particularly, ensure that all sheet edges are supported on structural framing with appropriate fixings used at the correct centres to avoid differential movement between adjacent sheets.
- All sheet edges shall be cut cleanly, with all excess debris and loose material appropriately removed prior to membrane application.
- Where required by the manufacturer of the sheeting material, ensure that an appropriately sized joint is integrated between adjacent sheets, with the joint being appropriately treated with an approved compound (typically, Dymonic 100).
- The membrane system shall be appropriately detailed across all sheet joints, using a combination of Perm-a-Fab, Polyflash Tape and/ or DualFlex Bandage. Consult with your local Tremco CPG representative for project specific advice.
- Undertake substrate specific moisture testing on the sheeting material, to ensure that it is sufficiently dry prior to priming and membrane application.

- Consideration should be given to the overall design of the structure, to mitigate against the potential for condensation to occur beneath, or rising vapour to affect the installed membrane.
- In line with regulatory requirements, Tremco CPG require that all sheeting materials are inherently water resistant, with all cut edges appropriately treated to maintain the inherent water resistance of the sheeting.
- It is not recommended to use particleboard sheeting as a substrate for waterproofing systems.
- Seek project specific advice from Tremco CPG, where the proposed lightweight sheeting material is treated with a pre-applied, film forming coating.

2.3. Substrate Preparation- Non-Porous Substrates

- Tremco CPG membrane and coating systems can typically be detailed onto small sections of various non-porous substrates, including, but not necessarily limited to; drainage outlets, pipe penetrations, balustrades, and in-situ hobs.
- All metal surfaces shall be mechanically abraded to meet the requirements in AS 1627.4, class 2.5 for “Near White Metal”.
- All plastic surfaces shall be mechanically abraded to create a profile to assist with the subsequent adhesion of the membrane/ coating system.
- All non-porous substrates are to be cleaned via an IPA wipe, using the 2-cloth method, ensuring that all residual solvent is allowed to flash off prior to priming.
- Particular attention should be paid where a coating/ membrane is to be installed over a galvanised substrate, as the zinc coating may prevent the system from achieving satisfactory long-term adhesion. Consult your local Tremco CPG representative for project specific advice.
- Notwithstanding the above, Tremco CPG recommends that project specific adhesion testing is undertaken on a representative sample to ensure that the level of preparation and priming allows the membrane system to achieve satisfactory adhesion over the non-porous substrate.
- Consult with your local Tremco CPG representative for project specific advice where it is proposed to apply the nominated system to a large area over a non-porous substrate, or over any other type of substrate.

2.4. Substrate Preparation- Blindside Membranes

2.4.1. Below Slab on Grade

- The sub-base should be prepared by either installing a blinding slab (preferred), compacting the original earth, or compacting a granular base to meet a minimum modified 85% proctor density per Engineer’s design.
- Irrespective of the sub-base preparation, the substrate should be free of angular projections and voids, consult your local Tremco CPG representative for project specific recommendations.
- All penetrations must be secured prior to detailing. For single pipe penetrations, refer to Tremco CPG Australia standard detail drawings.
- Multiple penetrations shall be spaced a minimum of 150mm apart to allow for proper detailing. If 150mm spacing is not achievable, contact your local Tremco CPG representative for a project specific recommendation.

2.4.2.Blindside Walls

- Tremco CPG blindside membranes can be applied to a variety of shoring wall structures, including but not necessarily limited to; secant piles, soldier piles and sheet piles.
- Irrespective of the type of shoring wall used, the wall substrate shall be appropriately prepared so that it is smooth and free from any significant voids of protrusions.
- Where secant piles or sheet piles are used, the ‘crotches’ between secant piles or voids between adjacent sheets should be appropriately treated to provide a uniform substrate for membrane application.

- Typically, Tremco CPG recommends a minimum 19mm thick, pressure treated plywood fixed to the shoring wall with the void behind the plywood filled with lean mix concrete, sand or aggregate. The correct plywood thickness and anchor spacings shall be determined by the engineer of record. Alternatively, polystyrene void formers may be used, subject to project specific approval from Tremco CPG.
- All penetrations must be secured prior to detailing. For single pipe penetrations, refer to Tremco CPG Australia standard detail drawings.
- Multiple penetrations shall be spaced a minimum of 150mm apart to allow for proper detailing. If 150mm spacing is not achievable, contact your local Tremco CPG representative for a project specific recommendation.

2.5. Crack and Joint Treatment

Unless otherwise noted on the product specific technical literature, cracks and joints in the substrate shall be treated in the following manner:

2.5.1. Liquid Applied Waterproofing Systems

Non-structural static cracks < 1.6mm wide

- A 150mm wide detail coat of membrane shall be installed over the primed crack prior to the installation of the complete waterproofing system.

Non-structural static cracks > 1.6mm wide and construction joints

- Non-structural static cracks that are 1.6mm wide or greater shall be ground out to a minimum 6mm wide, and subsequently treated with a compatible sealant.
- Ensure that backing rod or bond breaker tape is installed at the base of the joint to prevent 3-sided adhesion of the sealant.
- The depth of the grind shall be adjusted depending upon whether backing rod or bond breaker tape is to be used, to ensure that the sealant is able to be installed to a 1:1 (width: depth) ratio, assuming a 6mm joint width.
- The treated crack shall then be treated with a 150mm wide detail coat of membrane prior to the installation of the complete waterproofing system.

Live cracks and joints

- Seek project specific advice from your local Tremco CPG representative, as the way in which the crack/ joint will be treated may vary depending on the maximum anticipated movement, and the desired overburden finish.

2.5.2. Methyl Methacrylate Systems

Non-structural cracks between 0.5mm to 1.0mm wide

- A 200mm wide coat of Flowfast Primer shall be applied over the crack.
- Once the Flowfast Primer has cured, a 150mm detail coat of Deckshield PDS Membrane shall be applied, ensuring that it is centred over the crack.
- When applying the complete MMA coating to the remainder of the area, the coat of Flowfast Primer being applied to the main deck, must not be applied over the Deckshield PDS Membrane detail coat that has been applied to detail the cracks.

Non-structural cracks between 1.0mm to 1.5mm wide

- Non-structural cracks that are between 1.0mm to 1.5mm wide shall be ground out to a minimum width of 6mm and depth of 12mm.
- Flowfast Primer shall be applied to the vertical edges and extend approximately 100mm either side of the cut.
- Once the Flowfast Primer has cured, the cut shall be filled using the Deckshield PDS membrane.
- Once the Deckshield PDS membrane has cured, a min. 150mm wide reinforced detail coat of Deckshield PDS Membrane shall be applied, ensuring that it is centred over the crack. To reinforce the detail coat of Deckshield PDS membrane, Perm-a-Fab reinforcing fabric shall be

integrated into the Deckshield PDS membrane whilst it still wet, immediately followed by a second layer of Deckshield PDS Membrane, to encapsulate the reinforcing fabric.
- When applying the complete MMA coating to the remainder of the area, the coat of Flowfast Primer being applied to the main deck, must not be applied over the Deckshield PDS Membrane detail coat that has been applied to detail the cracks.

Construction joints
- Construction joint shall be ground out to a minimum width of 6mm and depth of 12mm.
- Flowfast Primer shall be applied to the vertical edges and extend approximately 100mm either side of the cut.
- Once the Flowfast Primer has cured, backing rod shall be installed to leave a 6mm x 6mm cut, which shall be filled using the Deckshield PDS membrane.
- Once the Deckshield PDS membrane has cured, a min. 150mm wide reinforced detail coat of Deckshield PDS Membrane shall be applied, ensuring that it is centred over the crack. To reinforce the detail coat of Deckshield PDS membrane, Perm-a-Fab reinforcing fabric shall be integrated into the Deckshield PDS membrane whilst it still wet, immediately followed by a second layer of Deckshield PDS Membrane, to encapsulate the reinforcing fabric.
- When applying the complete MMA coating to the remainder of the area, the coat of Flowfast Primer being applied to the main deck, must not be applied over the Deckshield PDS Membrane detail coat that has been applied to detail the cracks.

Live cracks and joints, and cracks > 1.5mm wide
- Seek project specific advice from your local Tremco CPG representative, as the way in which the crack/ joint will be treated may vary depending on the maximum anticipated movement, and the desired overburden finish.

2.5.3.Epoxy and Polyurethane Cement Systems
Non-structural static cracks
- Non-structural static cracks do not typically require additional treatment where a 100% solids epoxy primer is to be used, provided that no movement is anticipated across the cracks.

Live cracks and joints
- Seek project specific advice from your local Tremco CPG representative, as the way in which the crack/ joint will be treated may vary depending on the maximum anticipated movement, and the desired overburden finish.

2.6. Fillets and Coving
Unless otherwise noted on the product specific technical literature, the installation of fillets and coving associated with the membrane and/or protective coating system shall be undertaken in the following manner:

2.6.1.Waterproofing Systems
Internal Wet Area
- Joints, fillets, and bond breakers shall be installed as part of the internal wet area membrane system, in accordance with the information contained with AS 3740.
- Typically, where a Class III membrane system is to be installed, a 12mm bead of compatible sealant constitutes a suitable fillet.

External Above Grade
- Joints, fillets, and bond breakers shall be installed as part of the external above grade membrane system, in accordance with the information contained with AS 4654.2.
- Typically, where a liquid applied membrane system is to be installed, a 15mm x 15mm bead of compatible sealant constitutes a suitable fillet.

- Typically, where a sheet applied membrane system is to be installed, a 40mm x 40mm sand/cement cove constitutes a suitable fillet.
- All external corners shall be constructed with a chamfered edge to allow to the nominated membrane system to be installed to a consistent thickness across the corner.

Below Grade

- The installation of joints, fillets and coving associated with the below grade waterproofing membrane systems shall be undertaken in line with the information contained within the product specific technical literature.
- All external corners shall be constructed with a chamfered edge to allow the nominated membrane system to be installed to a consistent thickness across the corner.

2.6.2.Protective Coating Systems

- The coving associated with the installation of the protective coating system shall be undertaken using a compatible coving mortar. Refer to the information contained within the product specific technical literature for specific details.
- Alternatively, prefabricated coving profiles may be installed and sealed to the substrate with a compatible sealant.
- All external corners shall be constructed with a chamfered edge to allow to the nominated coating system to be installed to a consistent thickness across the corner.

2.7. Falls

- Ensure that all falls in the substrate are in accordance with the relevant regulatory requirements, particularly:
    - Internal Wet Areas- AS 3740
    - External Above Grade Areas - AS 4654.2
- In line with best practice, all substrates which are to be treated with a membrane system or protective coating should integrate falls to drainage outlets, even where surplus to regulatory requirements.
- The overall design should ensure that falls to drainage outlets are maintained, when considering the impact of long term (greater than 10 year) deflection of the structure.
- Euco Versaspeed LS100 may be used over a suitably prepared, reinforced concrete substrate to create falls prior to the installation of the nominated membrane system, consult with your local Tremco CPG representative for project specific advice.

2.8. Drains

- Ensure that all areas integrate sufficient drainage and overflow provisions, in accordance with the requirement of the Plumbing Code of Australia and AS 3500.
- All drainage outlets shall incorporate a horizontal surface to facilitate the appropriate termination of the membrane, and to mitigate against the potential for water to track beneath the installed system. Typically, either a leak control flange, or a plumbing fitting with an integral flange (siphonic outlet) are considered to be acceptable.
- It is imperative that the drainage system is designed and installed in a manner which is effective in allowing subsurface moisture to freely drain, the following items shall be noted in this regard:
    - The drainage flange shall be flush with (or preferably recessed into) the adjacent substrate.
    - Where dual membrane systems are adopted for usage (i.e. above and below screed), both the above screed and below screed membrane system should incorporate a flange to prevent moisture from becoming trapped within the screed.
- Where a membrane system is to be installed, a compatible connector sealant (as nominated on the product specific technical literature) shall be installed between the drainage flange and adjacent substrate.

3. Product Application

3.1. Extent of Application

- The extent of product application shall be in line with the information contained within:
  - Architectural design and specification.
  - Specialist consultant design and specification.
  - Regulatory requirements
- Where any contradictory requirements exist between the aforementioned items, the most stringent requirements shall be adopted.
- Tremco CPG offer a range of standard details which may be used by project design stakeholders as part of the project specific design development. These details are available via your local Tremco CPG representative or via www.tremcocpg.com.au.
- In addition, the Tremco CPG approved subcontractor shall use their skill, experience, and expertise to carry out any additional works that may be required to ensure a successful project outcome.

3.2. Material Storage

- All materials shall be stored in line with the information contained within the product specific technical literature, and work health and safety requirements.
- Unless otherwise noted, as a general rule all products shall be stored in unopened containers in a cool, dry location (between 5°C and 35°C and optimally between 12°C and 25°C) and conditioned to ambient conditions prior to product installation.

3.3. Product Mixing, Application and Finishing

- All materials shall be handled, mixed, installed and finished in line with the information contained within the product specific technical literature, and work health and safety requirements.
- Where plural component products are being used it is strongly recommended that full kits be mixed to ensure product homogeneity.
- Refer to the product specific technical literature for specific information regarding the impact that ambient conditions have on the installation of the nominated Tremco CPG system.
- Refer to the product specific technical literature for specific information regarding cleaning during the installation of the nominated membrane or protective coating system.
- As a general rule, material shall be cleaned using an approved solvent while in an uncured state. Cured material can typically only be removed by mechanical means.

3.4. Curing

- All installed materials shall be allowed to cure to the appropriate stage prior to subsequent works being undertaken. Refer to the product specific technical literature for further information.
- Consideration shall be given to the various stages of product curing:
  - Recoat window: The minimum to maximum period of time where a material can be recoated/ overcoated without reactivation being required.
  - Overlay stage: The period of time after which subsequent finishes (e.g. screed, topping, growing medium, tiles etc.) can be installed.
  - Full cure stage: The period of time after which the material has reached its specified physical properties, and can be placed into full service (including traffic and chemical exposure as applicable)
- It is important to note that ambient conditions may affect the rate of curing. Consult with your local Tremco CPG representative for project specific advice.

3.5. Termination and Penetrations

- All systems shall be appropriately terminated in line with the information contained within the project specific design documentation, but in any case, not less than the regulatory requirements, specifically AS 3740 for internal wet areas and AS 4654.2 for external above grade areas.

- Ensure that all systems are terminated at an appropriate location utilising a suitable means to prevent moisture tracking behind the installed system.
- Ensure that all fixings that penetrate the installed system are appropriately detailed to maintain the integrity of the installed system. As best practice, Tremco CPG suggests that fixings are installed using a chemical set epoxy to engineer's details, and subsequently over sealed with a compatible sealant.

3.6. Joints

- All joints shall be treated in line with the information contained within the project specific design documentation.
- Ensure that the proposed joint design is suitable to withstand the maximum anticipated movement across the joint, traffic, and chemical exposure (as applicable).
- Typically, where a conventional joint sealant is to be used, the joint shall be designed in accordance with the following criteria:
    - Joint width should be approximately 4 times the maximum anticipated movement of the joint (up to a width of approximately 25mm).
    - 1:1 (width: depth) ratio for joints greater than 6mm but less than 12mm wide.
    - 2:1 (width: depth) ratio for joints greater than 12mm wide but less than 25mm wide.
- Consult with your local Tremco CPG representative for project specific advice, which may vary depending on the maximum anticipated movement.

3.7. Protection of Installed Materials

- All materials shall be appropriately protected from damage during product installation, curing and service.
- Systems designed for non-UV exposure should typically be protected from UV within 4 weeks of product application. Consult with your local Tremco CPG representative for project specific advice, where this is not possible.
- Ensure that the installed system is appropriately protected prior to the installation of the subsequent finishes, depending on the area of application the following protection shall be integrated:
    - Planters/ Landscaped Areas: Protect system with a free drainage protection course (typically drainage cell and geofabric on horizontal surfaces, and dimpled protection board with integral geofabric on vertical surfaces) prior to the installation of the growing medium. Coreflute shall not be used. In line with best practice, only plants with non-invasive root systems should be used within planters and landscaped areas.
    - Pedestals/ Timber Decking: Protect system with geofabric or other rigid protection in local areas where pedestals or other structural supports are in direct contact with the installed system.
    - Brickwork: Protect system with geofabric or other rigid protection in local areas where brickwork is to be constructed in direct contact with the installed system.
    - Ballast: Protect system with a free drainage protection course (typically drainage cell and geofabric on horizontal surfaces, and dimpled protection board with integral geofabric on vertical surfaces) prior to the installation of the ballast. Coreflute shall not be used.
    - Reinforced concrete topping: Protect system with a double layer polyethylene plastic slip sheet prior to the installation of the reinforced concrete topping.
- Where the installed system is to have equipment installed directly above it (plant, solar panels, air conditioner units etc.), plinths or supports should be integrated to ensure that there is clearance between the equipment and the surface of the installed membrane, so that the ability of the area to freely drain is not impeded.

- Consult with CPG for project specific advice where the installed system is to be subject to; chemical exposure, significant direct traffic, extreme temperatures, or any other deteriorative element, to ensure that the materials warranty will be maintained, and to ensure that the proposed product remains fit for purpose.

3.8. Slip Resistance

- Where an exposed system is to be adopted, aggregate shall be integrated into the system to achieve the slip resistance nominated by the specifier (architect, engineer or consultant) within the project specific design documentation, ensuring compliance with regulatory requirements, specifically the NCC, HB 197 and AS 4586.
- As the slip resistance of a system is primarily a function of the aggregate broadcast, site specific slip resistance (incorporating accelerated wear) testing should be undertaken on a prototype area to ensure that the way in which the system is installed achieves the required slip resistance.

4. Quality Assurance

4.1. Inspection and Testing

Quality assurance inspection and testing shall be undertaken by the specialist subcontractor (or by a nominated third party) to meet project specific requirements. Typically, the following items shall be subject to inspection and testing:

- Substrate preparation.
    - Inspection to ensure that the substrate is free from mould, paint, sealers, coatings, curing agents, loose particles, and other contamination or foreign matter that may interfere with the adhesion.
    - Inspection to ensure that all substrates have been prepared to the appropriate surface profile, as applicable.
    - Testing to verify that the substrate moisture content is at a permissible level (either RH probe test to ASTM F2170 or surface moisture content test to ASTM F2659).
    - Confirmation that the substrate achieves the minimum required strength prior to coating or membrane application.
    - Inspection to ensure that all substrates to be treated with a membrane or coating system achieve the minimum required fall to the drainage point.
    - Where required, project specific adhesion testing should be undertaken on a representative sample to ensure that the level of preparation and priming allows the membrane system to achieve satisfactory adhesion over the substrate.
- Primer installation.
    - Inspection to ensure that the correct primer has been used for all substrates, including specialised primers for non-porous substrates as applicable.
    - Verification that primer recoat and overcoat windows have been adhered to prior to subsequent works commencing.
- Fillets, transitions and detail coats.
    - Inspection to ensure that all perpendicular junctions have been appropriately detailed, and an appropriately sized fillet has been installed.
    - Inspection to ensure that all joints, cracks and other critical junctions have been appropriately detailed.
    - Verification that the correct product has been used and has been allowed to sufficiently cure prior to subsequent works commencing.
- Membrane Installation.
    - Verification that the correct system has been installed, in accordance with the project specific design documentation and product specific technical literature.
    - Testing the wet film thickness (WFT) of the installed coating, as per the methodology outlined in ASTM D4414-95.

- Verification that membrane/ coating recoats and overcoat windows have been adhered to prior to subsequent works commencing.

- Post system application.
  - Visual Inspection to verify that the overall system has been installed in accordance with the information contained within the project specific design documentation and is free from any visual defects.
  - In addition to the visual inspection a combination of the following additional tests may be undertaken to satisfy project specific requirements:
    - Controlled water test (flood test) to be conducted for a minimum duration of 24 hours, and/or;
    - Electronic leak detection (either high voltage to AS 3894.1 or low voltage to ASTM D8231-19, to suit the system that has been installed, and/or;
    - Seam probe test (welded sheet membranes only), to identify any seams/laps that have not been appropriately welded (or alternatively undertake vacuum box testing for sheet membrane seams), and/or;
    - Non-destructive dry film thickness (DFT) testing, in accordance with AS 3894.3, and/or;
    - Adhesion testing, for trafficable exposed membrane systems, and/or;
    - Slip resistance and accelerated wear testing for trafficable exposed membrane systems.
- In addition to the above, contact Tremco CPG to undertake periodic iAuditor observations, which are designed to support the overall quality assurance program for the project.
  - A representative from Tremco CPG will attend site upon request (ideally to a pre-agreed inspection schedule or ad-hoc with appropriate notice provided) to observe aspects pertaining to the physical application of the nominated systems.
  - iAuditor observation reports document the observations made at the time of the site visit, as well as any corrective actions required by the installer to achieve a complete and warrantable system.

4.2. Maintenance

- All systems shall be maintained in strict accordance with Tremco CPG's published technical literature. The installed system shall be subject to regular inspection, with particular attention given to the following elements:
  - Areas exhibiting signs of excessive wear and/or mechanical damage.
  - Drainage points and overflows, to ensure that they remain functional.
  - Critical joints and junctions.
  - Areas subject to deteriorative elements (chemicals, extreme temperatures, heavy traffic, UV exposure etc.)
  - General cleaning and housekeeping.

5. Nominated Systems

Code	-
Area	Internal Wet Areas-Direct Stick Flooring
Component	**Details**
Primer (Porous Substrates)	TREMproof 200EC is a two-component water-based epoxy primer designed to act as a vapour retarder on porous substrates, to minimise vapour drive into liquid applied waterproofing membranes. TREMproof 200EC may also be applied to 'green' concrete, to act as an effective vapour retarder prior to the installation of Tremco liquid applied waterproofing membranes. TREMproof 200EC shall be applied at a rate of 5 m2/L to a min. 0.2 mm WFT to achieve a 0.09 mm DFT. Multiple coats of TREMproof 200EC shall be applied where substrate moisture content exceeds 4.5% or 75% RH.
Primer (Non-Porous Substrates)	TREMprime Non-Porous Primer is a one-part primer, used on metal or plastic substrates to enhance the adhesion of all Tremco polyurethane sealants and membranes. TREMprime Non-Porous Primer shall be applied at a rate of 40 m2/L to a min. 0.02 mm WFT to achieve a 0.01 mm DFT.
Coating/ Membrane System	***Above and Below Screed and Walls*** Vulkem H2 Plus is a single component, rapid cure, modified acrylic waterproofing membrane, incorporating micro fibres to increase strength and wet film thickness. Where being used as a waterproof membrane, as per the requirements of the NCC and relevant Australian Standard, Vulkem H2 Plus shall be applied at a rate of 1 m2/L to a min. 1.0 mm WFT over 2 coats to achieve a final 1.0mm DFT. Where being used as a water-resistant coating, surplus to the requirements of the NCC and relevant Australian Standard, Vulkem H2 Plus shall be applied at a rate of 1 m2/L to a min. 1.0 mm WFT to achieve a final 0.5 mm DFT.
Additional Products	Dymonic 100 is a high performance, best in class movement, medium-modulus, low VOC, UV stable, non-sag polyurethane sealant. Formulated with an innovative polymer technology, Dymonic 100 is a highly versatile sealant that has a unique capability to adhere to damp or green concrete. Used as a sealant fillet to detail floor/wall junctions, penetrations etc. TREMflex 50 is a low modulus joint sealant, especially formulated to ensure bubble free cure even at very high temperatures and humidity climatic conditions. It cures by reaction with atmospheric humidity to produce a joint sealant with a 50% joint movement accommodation factor. The sealant guns and tools consistently well at even the most extreme temperatures.

	VERSASPEED LS100 is a versatile, single component, rapid strength gaining repair mortar for horizontal and form and pour repair projects. Requiring only the addition of water, VERSASPEED LS100 is a high early strength material that is easy to use for fast turn-around projects. VERSASPEED LS100 is suitable for use in repairing concrete and creating falls from approximately 6mm to 150mm in thickness. Versaspeed LS100 may be installed over the prepared reinforced concrete substrate to create falls prior to the installation of the nominated membrane system, consult with Tremco CPG for project specific advice.
Specific Notes	Vulkem H2 Plus to be applied to a min 1.0mm DFT to all areas required to be waterproof as per AS 3740 requirements. Vulkem H2 Plus is a modified acrylic membrane and as such is typically compatible with most high-quality cementitious tile adhesives, notwithstanding this point we recommend that you seek project specific advice from Tremco CPG once the exact adhesive has been determined.
Materials Warranty	15-Year warranty

Code	-
Area	Window and Door Reveals
Component	**Details**
Primer (Porous Substrates)	TREMproof 200EC is a two-component water-based epoxy primer designed to act as a vapour retarder on porous substrates, to minimise vapour drive into liquid applied waterproofing membranes. TREMproof 200EC may also be applied to 'green' concrete, to act as an effective vapour retarder prior to the installation of Tremco liquid applied waterproofing membranes. TREMproof 200EC shall be applied at a rate of 5 m2/L to a min. 0.2 mm WFT to achieve a 0.09 mm DFT. Multiple coats of TREMproof 200EC shall be applied where substrate moisture content exceeds 4.5% or 75% RH.
Primer (Non-Porous Substrates)	TREMprime Non-Porous Primer is a one-part primer, used on metal or plastic substrates to enhance the adhesion of all Tremco polyurethane sealants and membranes. TREMprime Non-Porous Primer shall be applied at a rate of 40 m2/L to a min. 0.02 mm WFT to achieve a 0.01 mm DFT.
Coating/ Membrane System	Vulkem H2 Plus is a single component, rapid cure, modified acrylic waterproofing membrane, incorporating micro fibres to increase strength and wet film thickness. Vulkem H2 Plus shall be applied at a rate of 1 m2/L to a min. 1.0 mm WFT over 2 coats to achieve a final 1.0mm DFT. And (optional, only where required) TREMproof Façade is a high performance, water based acrylic membrane designed to provide a decorative and protective façade coating. TREMproof Façade is a single component, highly flexible, waterproof protective coating, which provides an aesthetically pleasing finish. TREMproof Facade shall be applied at a rate of 4.5 m2/L to a min. 0.215 mm WFT over 2 coats to achieve a final 0.25mm DFT.
Additional Products	Dymonic 100 is a high performance, best in class movement, medium-modulus, low VOC, UV stable, non-sag polyurethane sealant. Formulated with an innovative polymer technology, Dymonic 100 is a highly versatile sealant that has a unique capability to adhere to damp or green concrete. Used as a sealant fillet to detail floor/wall junctions, penetrations etc. PermAFab Reinforcing Fabric is a 100% stitch-bonded, polyester reinforcing fabric that offers a combination of high strength tensile properties, while also providing good elongation movement of the accompanying membrane for an overall excellent resistance to thermal stresses.

Specific Notes	Where the external walls are required to be waterproof to satisfy AS 4654.2 requirements, Tremco CPG recommends that Vulkem H2 Plus is installed, and subsequently overcoated with a UV protective coating to satisfy architectural aesthetic requirements (such as TREMproof Façade). Vulkem H2 Plus is an acrylic membrane and as such is typically compatible with most high-quality acrylic coatings and renders, notwithstanding this point we recommend that you seek project specific advice from Tremco once the exact detailing has been determined on site. Consult with Engineer/ Certifier regarding the non-combustibility requirements for these products. Where window reveals are directly adjacent to a external area that is to be waterproofed (as per AS 4654 requirements), the window reveal should be waterproofed with the membrane system nominated to treat the adjacent area, in lieu of the system nominated above.
Materials Warranty	10-years

Code	-
Area	External Deck Paver-Direct Stick
Component	**Details**
Primer (Porous Substrates)	TREMproof 200EC is a two-component water-based epoxy primer designed to act as a vapour retarder on porous substrates, to minimise vapour drive into liquid applied waterproofing membranes. TREMproof 200EC may also be applied to 'green' concrete, to act as an effective vapour retarder prior to the installation of Tremco liquid applied waterproofing membranes. TREMproof 200EC shall be applied at a rate of 5 m2/L to a min. 0.2 mm WFT to achieve a 0.09 mm DFT. Multiple coats of TREMproof 200EC shall be applied where substrate moisture content exceeds 4.5% or 75% RH.
Primer (Non-Porous Substrates)	TREMprime Non-Porous Primer is a one-part primer, used on metal or plastic substrates to enhance the adhesion of all Tremco polyurethane sealants and membranes. TREMprime Non-Porous Primer shall be applied at a rate of 40 m2/L to a min. 0.02 mm WFT to achieve a 0.01 mm DFT.
Coating/ Membrane System	Vulkem H2 Plus is a single component, rapid cure, modified acrylic waterproofing membrane, incorporating micro fibres to increase strength and wet film thickness. Vulkem H2 Plus shall be applied at a rate of 1 m2/L to a min. 1.0 mm WFT over 2 coats to achieve a final 1.0mm DFT.
Additional Products	Dymonic 100 is a high performance, best in class movement, medium-modulus, low VOC, UV stable, non-sag polyurethane sealant. Formulated with an innovative polymer technology, Dymonic 100 is a highly versatile sealant that has a unique capability to adhere to damp or green concrete. Used as a sealant fillet to detail floor/wall junctions, penetrations etc. TREMflex 50 is a low modulus joint sealant, especially formulated to ensure bubble free cure even at very high temperatures and humidity climatic conditions. It cures by reaction with atmospheric humidity to produce a joint sealant with a 50% joint movement accommodation factor. The sealant guns and tools consistently well at even the most extreme temperatures. VERSASPEED LS100 is a versatile, single component, rapid strength gaining repair mortar for horizontal and form and pour repair projects. Requiring only the addition of water, VERSASPEED LS100 is a high early strength material that is easy to use for fast turn-around projects. VERSASPEED LS100 is suitable for use in repairing concrete and creating falls from approximately 6mm to 150mm in thickness. Versaspeed LS100 may be installed over the prepared reinforced concrete substrate to create falls prior to the installation of the nominated membrane system, consult with Tremco CPG for project specific advice.
Specific Notes	

	Vulkem H2 Plus is a modified acrylic membrane and as such is typically compatible with most high-quality cementitious tile adhesives, notwithstanding this point we recommend that you seek project specific advice from Tremco CPG once the exact adhesive has been determined. Consideration should be given to subsurface moisture management considering both substrate falls and subsurface drainage provisions, to promote efficient water shedding from surface of the membrane. Select Tremco CPG waterproofing systems (excluding Flowcrete systems) may be applied over substrates constructed from lightweight sheeting. Consult with your local Tremco CPG representative for project specific advice. The surface to be coated must be dry, clean, smooth, firm, free of release agents, dust, mud, wires, fins, metal, projections, or any other substance that may prevent the nominated membrane system from achieving satisfactory adhesion. Ensure that the sheeting is appropriately installed in line with the manufacturer's and/or engineer's recommendations. Particularly, ensure that all sheet edges are supported on structural framing with appropriate fixings used at the correct centres to avoid differential movement between adjacent sheets. All sheet edges shall be cut cleanly, with all excess debris and loose material appropriately removed prior to membrane application. Where required by the manufacturer of the sheeting material, ensure that an appropriately sized joint is integrated between adjacent sheets, with the joint being appropriately treated with an approved compound (typically, Dymonic 100). The membrane system shall be appropriately detailed across all sheet joints, using a combination of Perm-a-Fab, Polyflash Tape and/ or DualFlex Bandage. Consult with your local Tremco CPG representative for project specific advice. Undertake substrate specific moisture testing on the sheeting material, to ensure that it is sufficiently dry prior to priming and membrane application. Consideration should be given to the overall design of the structure, to mitigate against the potential for condensation to occur beneath or rising vapour to affect the installed membrane. In line with regulatory requirements, Tremco CPG require that all sheeting materials are inherently water resistant, with all cut edges appropriately treated to maintain the inherent water resistance of the sheeting.

	It is not recommended to use particleboard sheeting as a substrate for waterproofing systems. Seek project specific advice from Tremco CPG, where the proposed lightweight sheeting material is treated with a pre-applied, film forming coating.
Materials Warranty	10-Year warranty

Code	-
Area	Other Additional Areas
Component	**Details**
-	***Fire Protection*** Nullifire Intumescent Coatings and Perlifoc HP Eco+ Fire Protection Mortars Consult with Chris Partington for project specific product recommendations as required (0498 002 633 and chris.partington@tremco.com.au) ***Curing Compound*** Evencure XDS-NXGEN is a hydrocarbon resin blend cure blended with hardeners and densifiers to create the highest performance curing compound currently available on the Australian market. This unique formulation offers many advantages over existing resin based curing compounds. All residual traces of curing compound to be removed from substrate prior to membrane application. Tremco CPG offer a complete range of Curing Compounds and Form Release Agents. ***Waterstops*** PVC Waterstop and Hydrophilic Waterstop to treat joints in the structure. Seek project specific advice from your local Tremco CPG representative. ***Jointing*** Willseal products specialise in innovative, pre-compressed, fire rated joint sealants that provide solutions for expansion and movement joints in almost every vertical and horizontal application. Seek project specific advice from your local Tremco CPG representative. ***Concrete Repair and Grouts*** Euclid Chemical is a world leading manufacturer of specialty chemical products for the concrete and masonry construction industry. Our local team of industry experts have created a selection of tailor-made concrete and masonry products for the local Australian construction industry. From Cementitious Fairing Coats and Flowable Repair Mortars to High Build and Structural Repair Mortars, our Eucocrete products will provide the solution for concrete structures across the Pacific. Seek project specific advice from your local Tremco CPG representative. ***Glazing and Facades*** Tremco CPG offer a range of innovative, commercial glazing and façade systems to treat the entire building envelope. Seek project specific advice from your local Tremco CPG representative.
Materials Warranty	TBC